

Frank — AMC Medal Drill

25 Jun 2026 | ANSWER KEY & method — for the parent

1. Answer: 6 [M5 · twin of real 2024 Q24 · 5 marks]

A box held 60 counters but some were lost. When the rest are shared equally among 4 people, 2 are left over; shared among 5 people, 4 are left over. Which of these could be the number lost: 4, 5, 6, 19, 23?

★ Fastest method: Bucket by remainder (or build an ordered table): count all-from-one-bucket PLUS one-from-each — no missing or double counts.

Trap to avoid: forgetting a case, or counting ordered when it should be unordered.

2. Answer: 126 [M2 · twin of real 2025 Q26 · 6 marks]

What is the smallest three-digit number that is even, a multiple of 6, contains the digit 2, has no digit 0, and has all different digits?

★ Fastest method: Apply the constraints in order of pruning power (divisibility / parity first) while keeping the leading digit as small as possible.

Trap to avoid: applying the weak constraints first, or dropping the distinct / no-zero filters.

3. Answer: 132 [M2 · twin of real 2025 Q26 · 6 marks]

What is the smallest three-digit number that is even, a multiple of 6, contains the digit 3, has no digit 0, and has all different digits?

★ Fastest method: Apply the constraints in order of pruning power (divisibility / parity first) while keeping the leading digit as small as possible.

Trap to avoid: applying the weak constraints first, or dropping the distinct / no-zero filters.

4. Answer: 152 [M2 · twin of real 2025 Q26 · 6 marks]

What is the smallest three-digit number that is even, a multiple of 4, contains the digit 5, has no digit 0, and has all different digits?

★ Fastest method: Apply the constraints in order of pruning power (divisibility / parity first) while keeping the leading digit as small as possible.

Trap to avoid: applying the weak constraints first, or dropping the distinct / no-zero filters.

5. Answer: 55 [M5 · twin of real 2013 Q27 · 7 marks]

In how many ways can you choose 3 different numbers from 2 to 13 so that their sum is a multiple of 4?

★ Fastest method: Bucket by remainder (or build an ordered table): count all-from-one-bucket PLUS one-from-each — no missing or double counts.

Trap to avoid: forgetting a case, or counting ordered when it should be unordered.

6. Answer: 110 [M5 · twin of real 2013 Q27 · 7 marks]

In how many ways can you choose 3 different numbers from 4 to 15 so that their sum is a multiple of 2?

★ Fastest method: Bucket by remainder (or build an ordered table): count all-from-one-bucket PLUS one-from-each — no missing or double counts.

Trap to avoid: forgetting a case, or counting ordered when it should be unordered.

Source-grounded: each item passed verify.py (unique answer + difficulty inside the real-paper band) and twins a specific real anchor. Focus areas chosen from Frank's 2018 misses (Q27 enumeration, Q30 cross-number).